

PURPLLE ONLINE BEAUTY SHOPPING

CHAPTER – 6

PRODUCT AND CATEGORY MANAGEMENT SYSTEM

1.Introduction

The Purplle Online Beauty Shopping Database Management System is designed to manage product and category information efficiently. The system uses two related tables, Category and Product, to store and organize beauty product details. Primary key and foreign key constraints are used to maintain data integrity. SQL operations such as insertion, updating, and deletion are performed on the Product table. Category-wise product reports are generated using SQL.

2.Objectives

- Design tables for Product and Category entities.
- Define primary and foreign key relationships.
- Store product name, category, price and stock.
- Store Product details such as Brand_ID, Seller_ID, Admin_ID, Description and Expiry_Date.
- Perform product insertion, updating and deletion.
- Generate category-wise product reports using SQL.

3.Table Design

3.1 Category Table

Attribute	Data Type	Key	Description
Category_ID	INT	Primary Key	Unique category identifier
Category_Name	VARCHAR2(100)	Unique	Beauty product category name
Description	VARCHAR2(255)	-	Category description

3.2 Product Table

Attribute	Data Type	Key	Description
Product_ID	INT	Primary Key	Unique product identifier
Product_Name	VARCHAR2(100)	-	Product name
Category_ID	INT	Foreign Key	References Category(Category_ID)
Brand_ID	INT	-	Brand reference
Seller_ID	INT	-	Seller reference
Admin_ID	INT	-	Admin reference
Description	VARCHAR2(255)	-	Product description
Expiry_Date	DATE	-	Product expiry date
Price	NUMBER(10,2)	-	Product price
Stock	INT	-	Available quantity

4. Primary and Foreign Key Relationship

Category → PurpleProduct

- **Category.Category_ID → Primary Key**
- **PurpleProduct.Product_ID → Primary Key**
- **PurpleProduct.Category_ID → Foreign Key referencing Category.Category_ID**
- **Relationship: One-to-Many (1:M)**
- **Meaning: One category can contain many products.**

CATEGORY (1) ————— (M) PURPLEPRODUCT
Category_ID (PK) Category_ID (FK)

5. Table Creation

5.1 Create Category Table

```
SQL> CREATE TABLE Category (  
2     Category_ID INT PRIMARY KEY,  
3     Category_Name VARCHAR2(50) NOT NULL UNIQUE,  
4     Description VARCHAR2(255)  
5 );
```

Table created.

5.2 Create Product Table

```
SQL> CREATE TABLE PurplleProduct (  
2     Product_ID INT PRIMARY KEY,  
3     Product_Name VARCHAR2(100) NOT NULL,  
4     Category_ID INT NOT NULL,  
5     Brand_ID INT,  
6     Seller_ID INT,  
7     Admin_ID INT,  
8     Description VARCHAR2(255),  
9     Expiry_Date DATE,  
10    Price NUMBER(10,2) NOT NULL CHECK (Price > 0),  
11    Stock INT NOT NULL CHECK (Stock >= 0),  
12    CONSTRAINT FK_Product_Category  
13    FOREIGN KEY (Category_ID)  
14    REFERENCES Category(Category_ID)  
15 );
```

Table created.

6. Data Insertion

6.1 Category Records

```
SQL> INSERT INTO Category VALUES (1, 'Makeup', 'Makeup products');  
1 row created.  
  
SQL> INSERT INTO Category VALUES (2, 'Skincare', 'Skin care products');  
1 row created.  
  
SQL> INSERT INTO Category VALUES (3, 'Haircare', 'Hair care products');  
1 row created.  
  
SQL> INSERT INTO Category VALUES (4, 'Fragrance', 'Perfume and fragrance products');  
1 row created.  
  
SQL> INSERT INTO Category VALUES (5, 'Bath and Body', 'Bath and body products');  
1 row created.  
  
SQL> SELECT * FROM Category;
```

6.2 Product Records

```
SQL> INSERT INTO PurplleProduct VALUES (101, 'Lakme Foundation', 1, 201, 301, 401, 'Liquid foundation', DATE '2027-05-10', 599, 50);
1 row created.

SQL> INSERT INTO PurplleProduct VALUES (102, 'Maybelline Lipstick', 1, 202, 302, 401, 'Matte lipstick', DATE '2027-06-15', 499, 40);
1 row created.

SQL> INSERT INTO PurplleProduct VALUES (103, 'Mamaearth Face Wash', 2, 203, 303, 402, 'Face cleansing product', DATE '2027-08-20', 349, 60);
1 row created.

SQL> INSERT INTO PurplleProduct VALUES (104, 'Minimalist Serum', 2, 204, 304, 402, 'Skin care serum', DATE '2027-09-10', 699, 35);
1 row created.

SQL> INSERT INTO PurplleProduct VALUES (105, 'Loreal Shampoo', 3, 205, 305, 403, 'Hair care shampoo', DATE '2028-01-15', 399, 45);
1 row created.

SQL> INSERT INTO PurplleProduct VALUES (106, 'Tresemmie Conditioner', 3, 206, 306, 403, 'Hair conditioner', DATE '2028-02-20', 449, 30);
1 row created.

SQL> INSERT INTO PurplleProduct VALUES (107, 'Fogg Perfume', 4, 207, 307, 404, 'Fragrance perfume', DATE '2029-03-10', 599, 25);
1 row created.

SQL> INSERT INTO PurplleProduct VALUES (108, 'Nivea Body Lotion', 5, 208, 308, 405, 'Moisturizing body lotion', DATE '2028-04-15', 299, 55);
1 row created.

SQL> INSERT INTO PurplleProduct VALUES (109, 'Dove Body Wash', 5, 209, 309, 405, 'Refreshing body wash', DATE '2028-05-20', 349, 40);
1 row created.

SQL> INSERT INTO PurplleProduct VALUES (110, 'Lakme Kajal', 1, 201, 301, 401, 'Eye makeup kajal', DATE '2027-07-10', 299, 70);
1 row created.
```

7. Data Retrieval

7.1 Display Category Records

```
SQL> SELECT * FROM Category;

CATEGORY_ID CATEGORY_NAME
-----
DESCRIPTION
-----
          1 Makeup
Makeup products

          2 Skincare
Skin care products

          3 Haircare
Hair care products

          4 Fragrance
Perfume and fragrance products

          5 Bath and Body
Bath and body products

SQL> COMMIT;

Commit complete.
```

7.2 Display Product Records

```

PRODUCT_ID
-----
PRODUCT_NAME
-----
CATEGORY_ID  BRAND_ID  SELLER_ID  ADMIN_ID
-----
DESCRIPTION
-----
EXPIRY_DA    PRICE    STOCK
-----
Matte lipstick
15-JUN-27      499      40

PRODUCT_ID
-----
PRODUCT_NAME
-----
CATEGORY_ID  BRAND_ID  SELLER_ID  ADMIN_ID
-----
DESCRIPTION
-----
EXPIRY_DA    PRICE    STOCK
-----
103
Mamaearth Face Wash
2          203      303      402

PRODUCT_ID
-----
PRODUCT_NAME
-----
CATEGORY_ID  BRAND_ID  SELLER_ID  ADMIN_ID
-----
DESCRIPTION
-----
EXPIRY_DA    PRICE    STOCK
-----
Face cleansing product
20-AUG-27    349      60

PRODUCT_ID
-----
PRODUCT_NAME
-----
CATEGORY_ID  BRAND_ID  SELLER_ID  ADMIN_ID
-----
DESCRIPTION
-----

```

```

SQL> SELECT * FROM PurpleProduct;

PRODUCT_ID
-----
PRODUCT_NAME
-----
CATEGORY_ID  BRAND_ID  SELLER_ID  ADMIN_ID
-----
DESCRIPTION
-----
EXPIRY_DA    PRICE    STOCK
-----
101
Lakme Foundation
1          201      301      401

PRODUCT_ID
-----
PRODUCT_NAME
-----
CATEGORY_ID  BRAND_ID  SELLER_ID  ADMIN_ID
-----
DESCRIPTION
-----
EXPIRY_DA    PRICE    STOCK
-----
Liquid foundation
10-MAY-27    599      50

PRODUCT_ID
-----
PRODUCT_NAME
-----
CATEGORY_ID  BRAND_ID  SELLER_ID  ADMIN_ID
-----
DESCRIPTION
-----
EXPIRY_DA    PRICE    STOCK
-----
102
Maybelline Lipstick
1          202      302      401

PRODUCT_ID
-----
PRODUCT_NAME
-----
CATEGORY_ID  BRAND_ID  SELLER_ID  ADMIN_ID
-----

```

```

DESCRIPTION
-----
EXPIRY_DA      PRICE      STOCK
-----
104
Minimalist Serum
2              204          304          402

PRODUCT_ID
-----
PRODUCT_NAME
-----
CATEGORY_ID    BRAND_ID    SELLER_ID    ADMIN_ID
-----
DESCRIPTION
-----
EXPIRY_DA      PRICE      STOCK
-----
Skin care serum
10-SEP-27      699          35

PRODUCT_ID
-----
PRODUCT_NAME
-----
CATEGORY_ID    BRAND_ID    SELLER_ID    ADMIN_ID
-----
DESCRIPTION
-----
EXPIRY_DA      PRICE      STOCK
-----
105
Loreal Shampoo
3              205          305          403

PRODUCT_ID
-----
PRODUCT_NAME
-----
CATEGORY_ID    BRAND_ID    SELLER_ID    ADMIN_ID
-----
DESCRIPTION
-----
EXPIRY_DA      PRICE      STOCK
-----
Hair care shampoo
15-JAN-28      399          45

```

```

PRODUCT_ID
-----
PRODUCT_NAME
-----
CATEGORY_ID    BRAND_ID    SELLER_ID    ADMIN_ID
-----
DESCRIPTION
-----
EXPIRY_DA      PRICE      STOCK
-----
106
Tresemme Conditioner
3              206          306          403

PRODUCT_ID
-----
PRODUCT_NAME
-----
CATEGORY_ID    BRAND_ID    SELLER_ID    ADMIN_ID
-----
DESCRIPTION
-----
EXPIRY_DA      PRICE      STOCK
-----
Hair conditioner
20-FEB-28      449          30

PRODUCT_ID
-----
PRODUCT_NAME
-----
CATEGORY_ID    BRAND_ID    SELLER_ID    ADMIN_ID
-----
DESCRIPTION
-----
EXPIRY_DA      PRICE      STOCK
-----
107
Fogg Perfume
4              207          307          404

PRODUCT_ID
-----
PRODUCT_NAME
-----
CATEGORY_ID    BRAND_ID    SELLER_ID    ADMIN_ID
-----
DESCRIPTION
-----

```

```

PRODUCT_ID
-----
PRODUCT_NAME
-----
CATEGORY_ID  BRAND_ID  SELLER_ID  ADMIN_ID
-----
DESCRIPTION
-----
EXPIRY_DA    PRICE    STOCK
-----
106
Tresemme Conditioner
3          206          306          403

PRODUCT_ID
-----
PRODUCT_NAME
-----
CATEGORY_ID  BRAND_ID  SELLER_ID  ADMIN_ID
-----
DESCRIPTION
-----
EXPIRY_DA    PRICE    STOCK
-----
Hair conditioner
20-FEB-28    449          30

PRODUCT_ID
-----
PRODUCT_NAME
-----
CATEGORY_ID  BRAND_ID  SELLER_ID  ADMIN_ID
-----
DESCRIPTION
-----
EXPIRY_DA    PRICE    STOCK
-----
107
Fogg Perfume
4          207          307          404

PRODUCT_ID
-----
PRODUCT_NAME
-----
CATEGORY_ID  BRAND_ID  SELLER_ID  ADMIN_ID
-----
DESCRIPTION
-----

```

```

PRODUCT_ID
-----
PRODUCT_NAME
-----
CATEGORY_ID  BRAND_ID  SELLER_ID  ADMIN_ID
-----
DESCRIPTION
-----
EXPIRY_DA    PRICE    STOCK
-----
106
Tresemme Conditioner
3          206          306          403

PRODUCT_ID
-----
PRODUCT_NAME
-----
CATEGORY_ID  BRAND_ID  SELLER_ID  ADMIN_ID
-----
DESCRIPTION
-----
EXPIRY_DA    PRICE    STOCK
-----
Hair conditioner
20-FEB-28    449          30

PRODUCT_ID
-----
PRODUCT_NAME
-----
CATEGORY_ID  BRAND_ID  SELLER_ID  ADMIN_ID
-----
DESCRIPTION
-----
EXPIRY_DA    PRICE    STOCK
-----
107
Fogg Perfume
4          207          307          404

PRODUCT_ID
-----
PRODUCT_NAME
-----
CATEGORY_ID  BRAND_ID  SELLER_ID  ADMIN_ID
-----
DESCRIPTION
-----

```

7. Product Update

```
SQL> UPDATE PurplleProduct SET Price = 649 WHERE Product_ID = 101;

1 row updated.

SQL> UPDATE PurplleProduct SET Price = 699, Stock = 60 WHERE Product_ID = 102;

1 row updated.

SQL> UPDATE PurplleProduct SET Price = Price + 50 WHERE Category_ID = 1;

3 rows updated.

SQL> SELECT * FROM PurplleProduct;
```

```
PRODUCT_ID
-----
PRODUCT_NAME
-----
CATEGORY_ID  BRAND_ID  SELLER_ID  ADMIN_ID
-----
DESCRIPTION
-----
EXPIRY_DA    PRICE    STOCK
-----
          101
Lakme Foundation
          1          201          301          401

PRODUCT_ID
-----
PRODUCT_NAME
-----
CATEGORY_ID  BRAND_ID  SELLER_ID  ADMIN_ID
-----
DESCRIPTION
-----
EXPIRY_DA    PRICE    STOCK
-----
Liquid foundation
10-MAY-27    699          50

PRODUCT_ID
-----
PRODUCT_NAME
-----
CATEGORY_ID  BRAND_ID  SELLER_ID  ADMIN_ID
-----
DESCRIPTION
-----
EXPIRY_DA    PRICE    STOCK
-----
          102
Maybelline Lipstick
          1          202          302          401

PRODUCT_ID
-----
PRODUCT_NAME
-----
CATEGORY_ID  BRAND_ID  SELLER_ID  ADMIN_ID
-----
DESCRIPTION
-----
EXPIRY_DA    PRICE    STOCK
-----
Matte lipstick
15-JUN-27    749          60
```

```

PRODUCT_ID
-----
PRODUCT_NAME
-----
CATEGORY_ID  BRAND_ID  SELLER_ID  ADMIN_ID
-----
DESCRIPTION
-----
EXPIRY_DA    PRICE    STOCK
-----
          110
Lakme Kajal
          1          201          301          401

PRODUCT_ID
-----
PRODUCT_NAME
-----
CATEGORY_ID  BRAND_ID  SELLER_ID  ADMIN_ID
-----
DESCRIPTION
-----
EXPIRY_DA    PRICE    STOCK
-----
Eye makeup kajal
10-JUL-27    349          70

```

8. Product Deletion

```

SQL> DELETE FROM PurplleProduct WHERE Product_ID = 110;

1 row deleted.

SQL> DELETE FROM PurplleProduct WHERE Product_Name = 'Fogg Perfume';

1 row deleted.

SQL> DELETE FROM PurplleProduct WHERE Stock = 0;

0 rows deleted.

SQL> DELETE FROM PurplleProduct WHERE Price < 300;

1 row deleted.

```

```

PRODUCT_ID
-----
PRODUCT_NAME
-----
CATEGORY_ID  BRAND_ID  SELLER_ID  ADMIN_ID
-----
DESCRIPTION
-----
EXPIRY_DA    PRICE    STOCK
-----
      101
Lakme Foundation
      1      201      301      401

PRODUCT_ID
-----
PRODUCT_NAME
-----
CATEGORY_ID  BRAND_ID  SELLER_ID  ADMIN_ID
-----
DESCRIPTION
-----
EXPIRY_DA    PRICE    STOCK
-----
Liquid foundation
10-MAY-27    699      50

PRODUCT_ID
-----
PRODUCT_NAME
-----
CATEGORY_ID  BRAND_ID  SELLER_ID  ADMIN_ID
-----
DESCRIPTION
-----
EXPIRY_DA    PRICE    STOCK
-----
      102
Maybelline Lipstick
      1      202      302      401

PRODUCT_ID
-----
PRODUCT_NAME
-----
CATEGORY_ID  BRAND_ID  SELLER_ID  ADMIN_ID
-----
DESCRIPTION
-----
EXPIRY_DA    PRICE    STOCK
-----
Matte lipstick
15-JUN-27    749      60

PRODUCT_ID
-----
PRODUCT_NAME
-----
CATEGORY_ID  BRAND_ID  SELLER_ID  ADMIN_ID
-----
DESCRIPTION
-----
EXPIRY_DA    PRICE    STOCK
-----
      103
Mamaearth Face Wash
      2      203      303      402

```

```
PRODUCT_ID
-----
PRODUCT_NAME
-----
CATEGORY_ID  BRAND_ID  SELLER_ID  ADMIN_ID
-----
DESCRIPTION
-----
EXPIRY_DA    PRICE    STOCK
-----
Face cleansing product
20-AUG-27    349      60
```

```
PRODUCT_ID
-----
PRODUCT_NAME
-----
CATEGORY_ID  BRAND_ID  SELLER_ID  ADMIN_ID
-----
DESCRIPTION
-----
EXPIRY_DA    PRICE    STOCK
-----
104
Minimalist Serum
2          204      304      402
```

```
PRODUCT_ID
-----
PRODUCT_NAME
-----
CATEGORY_ID  BRAND_ID  SELLER_ID  ADMIN_ID
-----
DESCRIPTION
-----
EXPIRY_DA    PRICE    STOCK
-----
Skin care serum
10-SEP-27    699      35
```

```
PRODUCT_ID
-----
PRODUCT_NAME
-----
CATEGORY_ID  BRAND_ID  SELLER_ID  ADMIN_ID
-----
DESCRIPTION
-----
EXPIRY_DA    PRICE    STOCK
-----
105
Loreal Shampoo
3          205      305      403
```

```
PRODUCT_ID
-----
PRODUCT_NAME
-----
CATEGORY_ID  BRAND_ID  SELLER_ID  ADMIN_ID
-----
DESCRIPTION
-----
EXPIRY_DA    PRICE    STOCK
-----
Hair care shampoo
15-JAN-28    399      45
```

```

PRODUCT_ID
-----
PRODUCT_NAME
-----
CATEGORY_ID  BRAND_ID  SELLER_ID  ADMIN_ID
-----
DESCRIPTION
-----
EXPIRY_DA      PRICE      STOCK
-----
      106
Tresemme Conditioner
      3      206      306      403

PRODUCT_ID
-----
PRODUCT_NAME
-----
CATEGORY_ID  BRAND_ID  SELLER_ID  ADMIN_ID
-----
DESCRIPTION
-----
EXPIRY_DA      PRICE      STOCK
-----
Hair conditioner
20-FEB-28      449      30

PRODUCT_ID
-----
PRODUCT_NAME
-----
CATEGORY_ID  BRAND_ID  SELLER_ID  ADMIN_ID
-----
DESCRIPTION
-----
EXPIRY_DA      PRICE      STOCK
-----
      109
Dove Body Wash
      5      209      309      405

PRODUCT_ID
-----
PRODUCT_NAME
-----
CATEGORY_ID  BRAND_ID  SELLER_ID  ADMIN_ID
-----
DESCRIPTION
-----
EXPIRY_DA      PRICE      STOCK
-----
Refreshing body wash
20-MAY-28      349      40

7 rows selected.

```

8. Category – wise Product Report

8.1 Product Listing by Category

```
SQL> SELECT
  2     C.Category_Name,
  3     P.Product_ID,
  4     P.Product_Name,
  5     P.Price,
  6     P.Stock
  7 FROM Category C
  8 JOIN PurpleProduct P
  9 ON C.Category_ID = P.Category_ID
 10 ORDER BY C.Category_Name;
```

CATEGORY_NAME	PRODUCT_ID

PRODUCT_NAME	

PRICE	STOCK

Bath and Body	109
Dove Body Wash	
349	40

Haircare	106
Tresemme Conditioner	
449	30

CATEGORY_NAME	PRODUCT_ID

PRODUCT_NAME	

PRICE	STOCK

Haircare	105
Loreal Shampoo	
399	45

Makeup	102
Maybelline Lipstick	

CATEGORY_NAME	PRODUCT_ID

PRODUCT_NAME	

PRICE	STOCK

749	60

Makeup	101
Lakme Foundation	
699	50

Skincare	103

CATEGORY_NAME	PRODUCT_ID

PRODUCT_NAME	

PRICE	STOCK

Mamaearth Face Wash	
349	60

Skincare	104
Minimalist Serum	
699	35

7 rows selected.

8.2 Products in Makeup Products Category

```
SQL> SELECT
  2     P.Product_ID,
  3     P.Product_Name,
  4     P.Price,
  5     P.Stock
  6 FROM PurpleProduct P
  7 JOIN Category C
  8 ON P.Category_ID = C.Category_ID
  9 WHERE C.Category_Name = 'Makeup products';
```

no rows selected

8.3 Products in Skincare Products Category

```
SQL> SELECT
  2     P.Product_ID,
  3     P.Product_Name,
  4     P.Price,
  5     P.Stock
  6 FROM PurpleProduct P
  7 JOIN Category C
  8 ON P.Category_ID = C.Category_ID
  9 WHERE C.Category_Name = 'Skincare';
```

PRODUCT_ID	PRODUCT_NAME	PRICE	STOCK
103	Mamaearth Face Wash	349	60
104	Minimalist Serum	699	35

8.4 Products in Haircare Products Category

```
SQL> SELECT
  2     P.Product_ID,
  3     P.Product_Name,
  4     P.Price,
  5     P.Stock
  6 FROM PurpleProduct P
  7 JOIN Category C
  8 ON P.Category_ID = C.Category_ID
  9 WHERE C.Category_Name = 'Haircare';
```

PRODUCT_ID	PRODUCT_NAME	PRICE	STOCK
105	Loreal Shampoo	399	45
106	Tresemme Conditioner	449	30

8.5 Products Count by Category

```
SQL> SELECT
2     C.Category_Name,
3     COUNT(P.Product_ID) AS Product_Count
4 FROM Category C
5 LEFT JOIN PurpleProduct P
6 ON C.Category_ID = P.Category_ID
7 GROUP BY C.Category_Name
8 ORDER BY C.Category_Name;
```

CATEGORY_NAME	PRODUCT_COUNT
Bath and Body	1
Fragrance	0
Haircare	2
Makeup	2
Skincare	2

8.6 Category-wise Total Stock

```
SQL> SELECT
2     C.Category_Name,
3     SUM(P.Stock) AS Total_Stock
4 FROM Category C
5 JOIN PurpleProduct P
6 ON C.Category_ID = P.Category_ID
7 GROUP BY C.Category_Name
8 ORDER BY C.Category_Name;
```

CATEGORY_NAME	TOTAL_STOCK
Bath and Body	40
Haircare	75
Makeup	110
Skincare	95

8.7 Category-wise Average Price

```
SQL> SELECT
2     C.Category_Name,
3     AVG(P.Price) AS Average_Price
4 FROM Category C
5 JOIN PurpleProduct P
6 ON C.Category_ID = P.Category_ID
7 GROUP BY C.Category_Name
8 ORDER BY C.Category_Name;
```

CATEGORY_NAME	AVERAGE_PRICE
Bath and Body	349
Haircare	424
Makeup	724
Skincare	524

Conclusion

The Product and Category database design successfully represents the one-to-many (1:M) relationship between categories and products in the Purple Online Beauty Shopping system. Primary and foreign key constraints maintain data integrity, while INSERT, UPDATE and DELETE operations support product management. The category-wise reports provide a useful summary of products available under each beauty category.